Experiment 4: Dense Map Reconstruction and Map-Quality Evaluation

Raw write-up for Chapter 4, Section 4.4

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Purpose of this document. This is the raw, complete write-up of Experiment 4, written before the thesis chapter; Section 4.4 of the final report is extracted from it. Every number traces to a file under `output/`, `map/` or `dataset/` named in the "Reporting rules" paragraph of its slot. Open concerns are set in red where they arise.

4.4.1 Simulation

Status: partial. Five databases are scored for geometry and drivability: the tiny warehouse, which passes, and four realistic-warehouse maps — two mapping recordings, one per layout, each built closure-free under the October and the current November configuration — all of which fail the quality criterion fixed before scoring. The tiny map's independent drivability reference remains lost.

Objective. Verify that the reconstructed map reflects the physical environment and is usable by a route planner.

Process. RTAB-Map 0.23.7 built a database from a mapping run. The occupancy grid was seated in world coordinates by one rigid transform measured from the map's own occupied cells against the world file, without odometry or ground truth. Geometry was then scored against the world's collision meshes, and drivability by stamping the true nav2 footprint along a ground-truth route.

Datasets. Table 4.1 lists the tiny-warehouse recordings and table 4.2 the realistic-warehouse ones. The tiny map's drivability reference is a separate run that no longer exists in either storage location; the only surviving recording of the tiny warehouse is the one that built the map. Each realistic map is probed with a recording that did not build it.

Table 4.1: Recordings of Experiment 4. Durations are rosbag wall-clock recording durations from the dataset registry; the simulator ran below real time.

Recording	World	Dur. (s)	Messages	Role
`dataset_map_tiny`	tiny, static	63.3	82 555	built `map/map_tiny.db`; substitute drivability route
`dataset_static_tiny_repeat`	tiny, static	–	2973 poses	independent drivability route; **lost**
`dataset_map_01`	large, static	193.5	152 291	large-warehouse mapping run; unscored

Table 4.2: Realistic-warehouse recordings of Experiment 4, all prefixed `dataset_allsensor_`. Build recordings are on the removable drive; their span is the stamp range of the stored nodes. Probe recordings are the static twins listed in the recordings table of the realistic-world note, read from the bags. U and I are the two layouts described there and in Chapter 3.

Recording	World	Span (s)	Truth	Role
`realistic_map`	U, static	285.7	14 411	formerly `_mapv2`; built U-Oct and U-Nov
`realistic_01_map`	I, static	314.9	15 760	built I-Oct and I-Nov
`realistic_000`	U, static	125.5	6 278	independent probe, 171.3 m
`realistic_001`	I, static	96.9	4 847	independent probe, 133.0 m

Reporting rules. The map under test is `map/map_tiny.db`, created 8 September 2026 from `small_warehouse_static/small_warehouse_static_tinymap.world`. The reference occupancy was rasterised from the world file's collision geometry at 0.05 m per cell; the fitted placement is $(1.850, 8.600)$ m at $-89.90°$, measured once by `script/fit_map_pose.py`. Occupied-cell precision is the share of occupied map cells within 0.25 m of true structure; recall is the share of true *surface* cells within tolerance of an occupied cell, over the region the map observed and over the whole building; drivability stamps the 0.83×0.50 m footprint from `nav2_ackermann.yaml` at every ground-truth pose and counts collisions (false obstacles), unknown cells and clearance. All of it is produced by `script/validate_map.py` into `output/compare/rtab_map_vs_gt.png` (independent route) and `output/compare/rtab_map_vs_gt_selfcheck.png` (build run). The experimental unit is one database, $n = 1$.

Results. The tiny map is geometrically faithful: 96.6 % of the cells it marks occupied lie on real structure. Table 4.3 gives the three geometry scores and their populations; table 4.4 gives drivability against the lost independent route and against the build run; table 4.5 summarises the stored graph. Figure 4.1 is the evaluation overlay for the independent route and figure 4.2 the same database against its own build run.

Table 4.3: Occupied-cell geometry of `map/map_tiny.db` against `small_warehouse_static_tinymap.world` at 0.25 m tolerance.

Metric	Value	Population
Occupied-cell precision	96.6 %	11,634 occupied cells
Recall, region the map observed	89.5 %	observed surfaces
Recall, whole building	67.2 %	all surfaces in the world file

Table 4.4: Drivability of `map/map_tiny.db` under the true 0.83×0.50 m nav2 footprint, against two routes. "Independent" is `dataset_static_tiny_repeat`, which did not build the map and is lost. "Build run" is `dataset/dataset_map_tiny`, the run the map was built from. Geometry is identical in both cases because it does not depend on the route.

Metric	Independent	Build run	Meaning
Poses scored	2973	3032	
Clear and mapped	99.3 %	100.0 %	footprint free and observed
False obstacle	0.7 %	0.0 %	proven error
Unmapped under footprint	0.0 %	0.0 %	blind cells on the route
Clearance, median	1.07 m	1.01 m	to nearest obstacle
Clearance, 5th percentile	0.45 m	0.46 m	inflation radius is 0.80 m

Analysis. The gap between observed-surface recall and whole-building recall is route coverage, not mapping failure. The run never entered the side aisles, so about a third of the building was never seen by

Table 4.5: Stored contents of `map/map_tiny.db`, read in read-only mode. Link rows are stored per direction; the undirected count is the number of distinct node pairs.

Element	Rows	Undirected	Note
Nodes	30	–	ids 1–30, stamp span 59.3 s
Neighbour links	56	28	sequential odometry constraints
Global loop closures	5	3	pairs $(9,22)$, $(10,23)$, $(29,30)$
Gravity links	30	30	one per node
Local-space links	0	0	none recorded
Visual words	10,357	–	vocabulary size
Features	28,374	–	stored keypoints

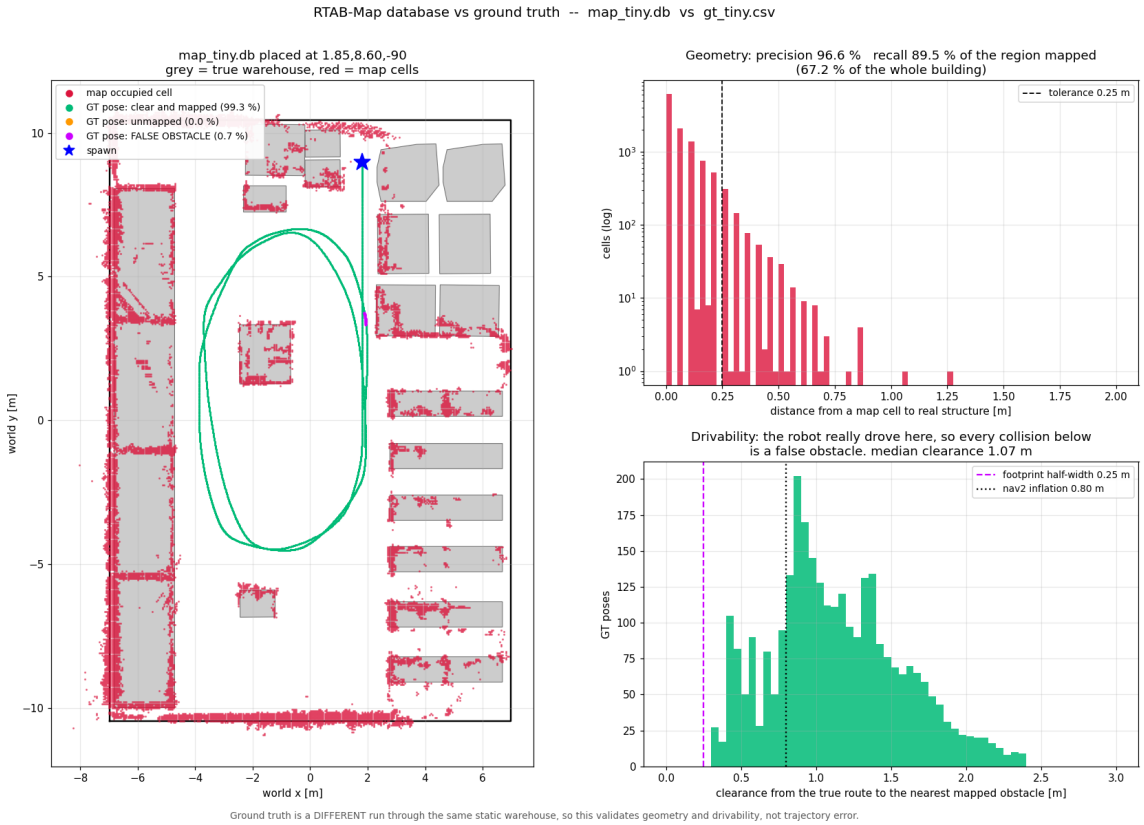


Figure 4.1: Dense-map evaluation of `map/map_tiny.db` against the independent route. Left: occupied cells (red) over the world file's collision geometry (grey), with the ground-truth route coloured by drivability and the spawn marked. Top right: distance from each occupied cell to real structure, with the 0.25 m tolerance marked. Bottom right: clearance from the route to the nearest mapped obstacle, against the footprint half-width and the nav2 inflation radius.

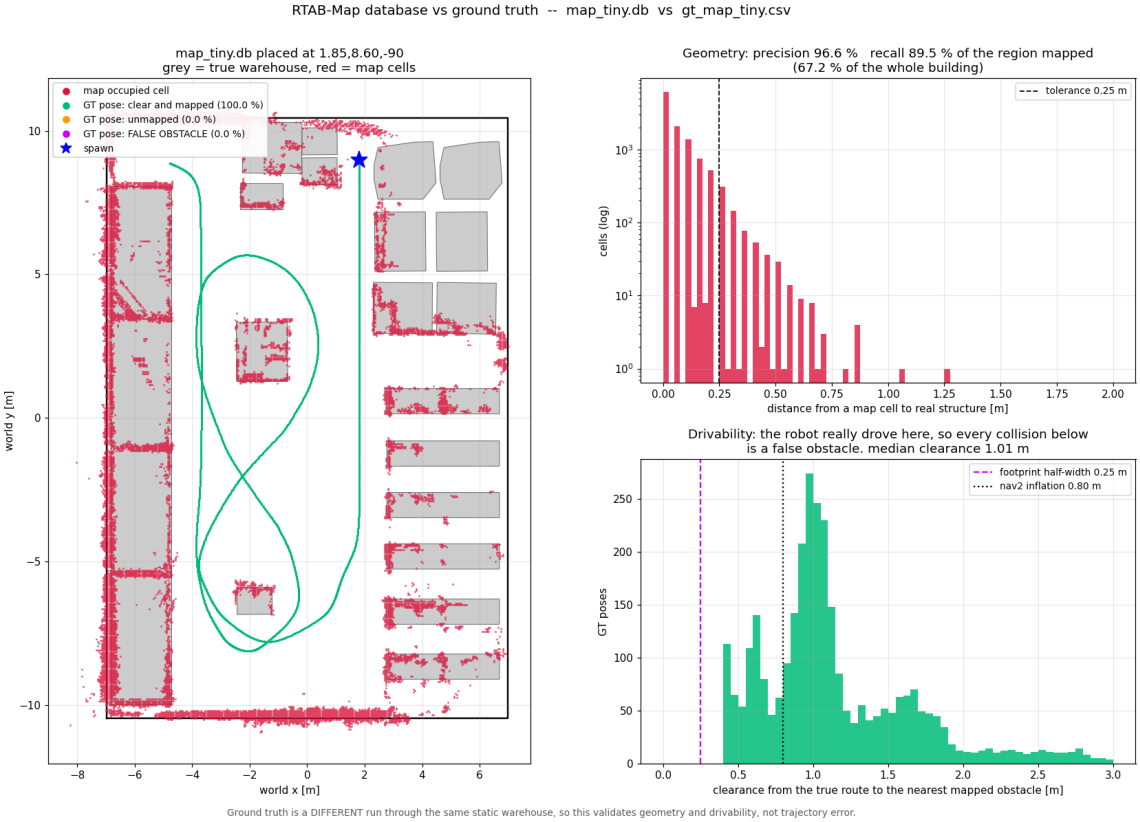


Figure 4.2: The same database scored against the run that built it, `dataset/dataset_map_tiny`. The route is a figure-eight rather than the oval of figure 4.1, and no pose on it collides with the map. The geometry panels are identical to figure 4.1. The footer claiming a different run is emitted unconditionally by the evaluator and is wrong for this figure.

any sensor; the first number describes mapping quality, the second where the robot went.

Placement is a measurement, not a tuned parameter. An independent prediction from the spawn pose of the IMU body frame gives $(1.800, 8.662)$ m, 0.080 m from the fitted value, and the fit beats the best rival orientation more than 5° away by a 60.9 % margin. Earlier assessments that judged this database unnavigable were made with the grid mis-seated, which is why placement is reported before any metric.

The two drivability columns measure the independence effect. Against the run that built it the map records no false obstacle at all, because it cannot contradict the trajectory it was constructed from; the 0.7 % that the independent route exposes is exactly what only an unseen route can reveal. The 0.0 % is a weaker question, not a better result, and is not a map-quality figure. Three measurements show the build run is the same run: its ground-truth poses at the stored node timestamps lie a median 0.340 m and at most 0.52 m from the nodes, the nodes lie a median 0.069 m from the route in space, and all thirty nodes fall inside the recording's sim-time interval.

Place recognition fired, but on a small graph. Two of the three closure pairs, $(9, 22)$ and $(10, 23)$, join nodes about half a lap apart and are genuine revisits; $(29, 30)$ joins consecutive nodes and is not evidence of place recognition.

Clearance couples to planner cost. The 5th-percentile clearance of 0.45 m sits below nav2's 0.80 m inflation radius on both routes, so nothing is blocked, but a planner would charge a high cost where the robot demonstrably drove.

Limitations, each stated once. *Lost reference:* the independent drivability route cannot be re-read, so that half of the result is a dated measurement. *Easy route:* one loop down the wide central aisle, so 0.7 % false obstacles is a floor, not a worst case. $n = 1$*:* one database on one world supports no uncertainty estimate and no claim about the large warehouse. *Residual distortion:* walls appear doubled and thickened and shelf rows bowed in the overlay; no rigid transform removes that, and it is the property that would limit navigation.

[Queried: reproducibility of the drivability half] The geometry result is reproducible: database and world file are both present. The drivability result is not: `dataset_static_tiny_repeat` is absent from `dataset/` and from the removable drive, the database stores an empty parameter set so its build configuration is unrecoverable, and the RTAB-Map engineering record's claim that the node stamps match no dataset is contradicted by the pose agreement above. The drivability figures stand as a dated measurement until an independent route is recorded and retained.

Large-warehouse attempt. The attempt **failed** before the back end was exercised. The progress update of 23 September 2026 records that the front end lost tracking in the repetitive bay structure of the large world, the estimated pose diverged from the ground-truth route within a few time steps of the loss, and the resulting map was not scorable; the cause was attributed to the front end under repeated structure, not to RTAB-Map. The mapping recording `dataset_map_01` survives in the registry. No run directory, placement fit, score or figure for this attempt is registered, so the result rests on the progress record alone and cannot be audited.

[Resolved and retired: unregistered databases] Four further databases sat beside the tiny map until 6 November 2026 (`map_01.db`, `map_02.db`, `map_02_vins.db`, `map_03.db`, all written on 21 September). The insight analysis of 22 October identified them from their contents: two aborted two-node starts, a second tiny-warehouse map on a different route, and an 18-node run that left the floor along one aisle and fits no large world, the large-warehouse attempt. All four were deleted on 6 November with the dataset decision, unplaced and unscored, so the attempt is again without an artefact.

Realistic-warehouse maps. Four databases built in the realistic worlds of Chapter 3 replace the unauditable large-warehouse attempt: two mapping recordings, one per layout, each built twice, on 26 October 2026 and on 6 November 2026 under the pipeline's current configuration. Every one has a

registered build recording, a fitted placement, a score and a figure, and all four fail the quality criterion fixed before they were scored.

Reporting rules. The criterion, fixed in `script/eval_realistic_maps.py` before any map was placed, is occupied-cell precision $\geq 90\,\%$, observed-region recall $\geq 85\,\%$ and false obstacles $\leq 1\,\%$ of probe poses, each taken from the tiny map's result; whole-building recall is reported, not tested. Placement was measured by `script/fit_map_pose.py` with the realistic spawn $(5.250, 19.995, -90^\circ)$ and confirmed on the overlays before scoring. Scoring is the unchanged `script/validate_map.py` at 0.25 m tolerance with the 0.83×0.50 m footprint, each map probed by the static twin of its own layout, a recording that built none of them. All four databases are closure-free (`Rtabmap/LoopThr` 1.0, `RGBD/ProximityBySpace` false; `loop_closure:=false`, RTAB-Map record §7c) and optimised with GTSAM. The two builds of each recording differ in exactly three back-end parameters, `Rtabmap/DetectionRate` 4 against 1 Hz, `Grid/RangeMax` 8 against 10 m and `Grid/MaxGroundHeight` 0.29 against 0.10 m (October against November), and in front-end session, which neither database records; they are two configurations, not a repeat pair, so $n = 1$ per cell and no difference between builds is attributable to one parameter. The recording now named `dataset_allsensor_realistic_map` is the one held until 26 October as `_mapv2` (identical message counts); the original 103–316 s mapping recording and the 114-node database built from it were deleted that day, and `map/map_realistic` is the former `map_realistic_v2`. Node stamps fix the pairing: the U databases span 13–300 s and the I databases 20–335 s, which the 125 s and 97 s probes cannot have produced. The November U placement was fitted in the -180° basin with `-yaw-near`, because the global winner, 4.2 % better by agreement, was the rotational rival the overlay rejects; both candidates are in `fit_dataset_allsensor_realistic_map.log`. Every number is under the experiment directory `exp4-realistic-maps` of `output/compare/simulation`: the scores in `realistic_map_metrics.csv`, one `validate_*.json` per map, the inventory with the three parameters in `map_inventory.json`, and the placement logs.

Results. Table 4.6 gives geometry, table 4.7 drivability against the independent probes, and table 4.8 the stored graphs and the three parameters that separate the builds. Figures 4.3 and 4.4 are the placement overlays of the two U builds and figures 4.5 to 4.8 the evaluation overlays.

Table 4.6: Occupied-cell geometry of the four realistic-warehouse databases against their world files at 0.25 m tolerance, beside the tiny map. Bold marks a value below the fixed criterion.

Database	Built	Nodes	Occupied cells	Precision [%]	Recall observed [%]	Recall building [%]
`map_tiny`	Sep	30	11 634	96.6	89.5	67.2
`map_realistic`	Oct	165	26728	**81.2**	**82.5**	55.7
`dataset_allsensor_realistic_map`	Nov	148	22715	**77.8**	**74.9**	57.1
`map_realistic_01`	Oct	185	25440	**84.8**	**81.4**	54.5
`dataset_allsensor_realistic_01_map`	Nov	127	23255	**82.2**	**77.1**	55.3

Analysis. All four maps fail, and the November build of each recording scores below its October sibling on every criterion: precision by 3.4 and 2.6 points, observed recall by 7.6 and 4.3, false obstacles from 3.8 to 5.0 and from 4.4 to 10.2 %. The October pair fail by similar margins on both layouts, which made their failure a property of the build rather than of one run; the November pair move it further in the same direction.

Three parameters moved together between the builds, and each has its own signature in the data, so the reading is consistent-with, not attribution. The detection rate, 4 to 1 Hz, shows as fewer nodes (148 against 165, 127 against 185) and fewer observations per cell, which the view-count effect of the tiny-map insight analysis turns into lower precision and recall. The grid range, 8 to 10 m, shows as structure the October builds never reached: the west wall and the far outer bays now carry occupied cells, and the

Table 4.7: Drivability of the four realistic-warehouse databases under the true 0.83×0.50 m nav2 footprint, against the independent probe of each layout; U-Oct is `map_realistic`, U-Nov `dataset_allsensor_realistic_map`, I-Oct `map_realistic_01`, I-Nov `dataset_allsensor_realistic_01_map`. The tiny map's independent figures were 2 973 poses, 0.7 % false obstacles, clearance 1.07 and 0.45 m; its probe was one loop of a wide aisle, so the comparison is not route-matched.

Metric	U-Oct	U-Nov	I-Oct	I-Nov
Probe poses	6278	6278	4847	4847
Clear and mapped [%]	96.2	95.0	94.9	89.5
False obstacle [%]	**3.8**	**5.0**	**4.4**	**10.2**
Unmapped under footprint [%]	2.4	1.9	0.7	0.5
Clearance, median [m]	1.96	2.00	2.16	1.98
Clearance, 5th percentile [m]	0.62	0.38	0.36	0.18

Table 4.8: Stored contents and build parameters of the four realistic-warehouse databases, read in read-only mode. The three parameter rows are the only ones that differ between the builds of a recording. Neighbour links are stored per direction; a "spanning" closure joins nodes more than three apart and none exists, by configuration. The margin is the agreement over the best rival more than 5° away; "basin" marks the placement chosen with `-yaw-near`.

	U-Oct	U-Nov	I-Oct	I-Nov
Built	2026-10-26	2026-11-06	2026-10-26	2026-11-06
`Rtabmap/DetectionRate` [Hz]	4.0	1.0	4.0	1.0
`Grid/RangeMax` [m]	8.0	10.0	8.0	10.0
`Grid/MaxGroundHeight` [m]	0.29	0.1	0.29	0.1
Nodes	165	148	185	127
Node stamp span [s]	285.7	286.2	314.9	313.9
Optimised path [m]	369.5	364.2	384.7	372.3
Neighbour links, rows	322	294	356	248
Closure rows / spanning > 3	3 / 0	0 / 0	4 / 0	0 / 0
Visual words	55802	51539	63276	44342
Database size [MB]	417	462	438	377
Placement (x, y, θ)	$(5.300, 19.500, -179.9^\circ)$	$(5.200, 19.800, -180.1^\circ)$	$(5.150, 20.450, 179.4^\circ)$	$(4.950, 20.400, -179.9^\circ)$
Fit agreement, margin	0.183, 10.1 %	0.161, –4.2 % (basin)	0.260, 66.1 %	0.252, 60.2 %

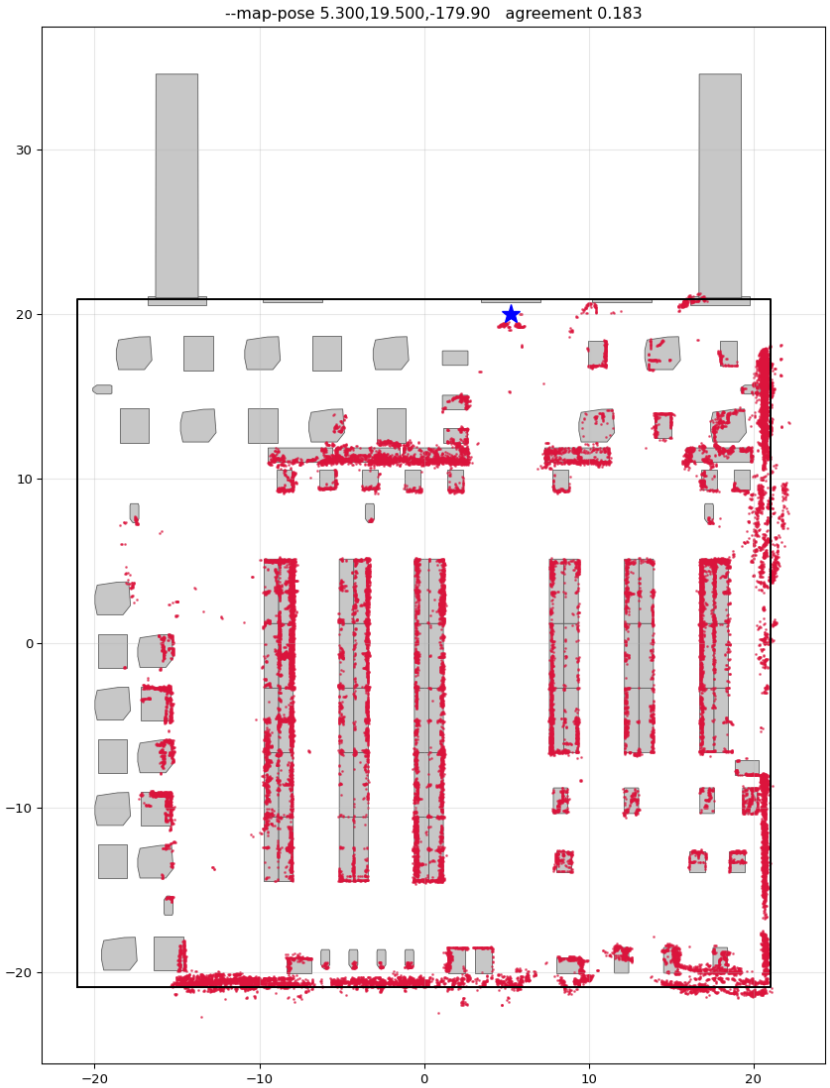


Figure 4.3: Placement overlay of `map/map_realistic` (U-Oct), agreement 0.183, 10.1 % margin over the rival orientation. Red occupied cells over the world file's obstacle footprints and wall; shelf rows, the bay strip at $y \approx 11$ and all four walls coincide.

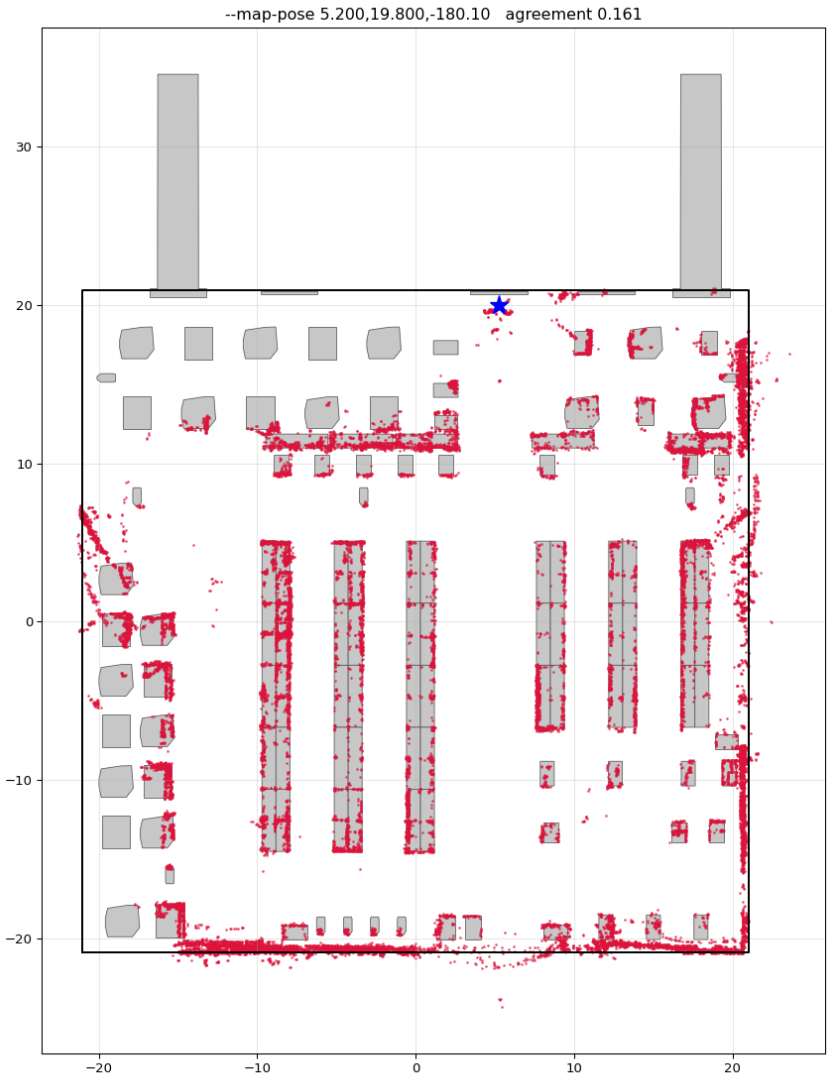


Figure 4.4: Placement overlay of `map/dataset_allsensor_realistic_map` (U-Nov). The unconstrained sweep preferred the 180°-rotated rival at $(-5.6, -19.6)$, agreement 0.168 against 0.161, which puts shelf rows between shelves; the basin placement shown sits 0.15 m from the spawn of the IMU body, as its October sibling does.

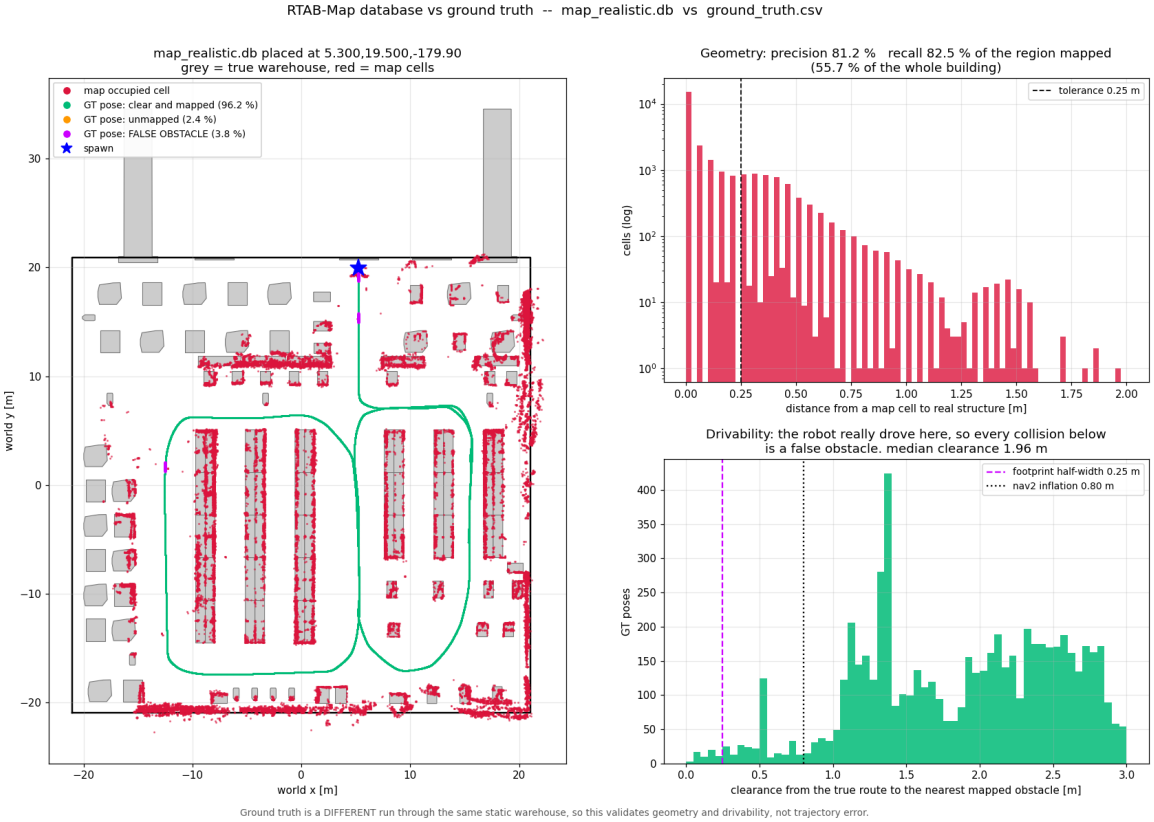


Figure 4.5: Dense-map evaluation of `map/map_realistic` (U-Oct) against `dataset_allsensor_realistic_000`. Panels as in figure 4.1. The shelf rows are thickened toward the aisles; the false-obstacle poses sit on the first metres below the spawn and where the probe hugs the left shelf column at $x \approx -12.5$.

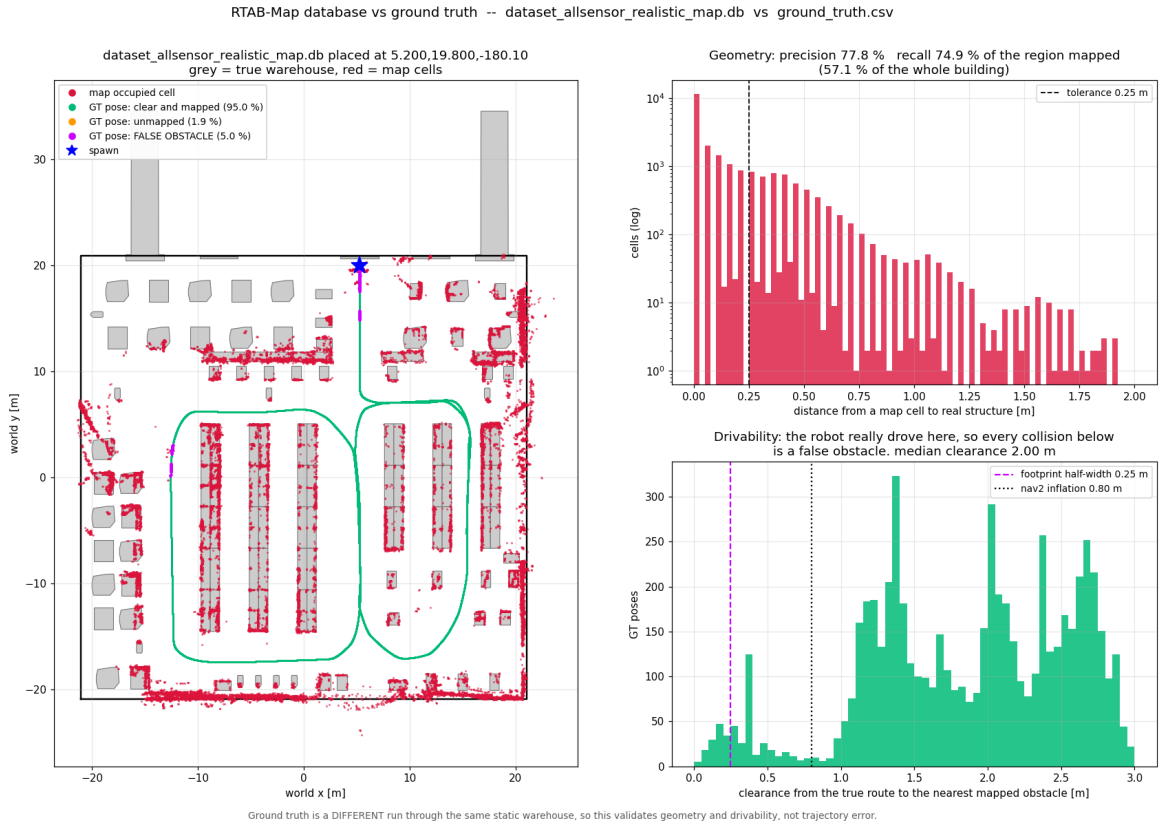


Figure 4.6: Dense-map evaluation of `map/dataset_allsensor_realistic_map` (U-Nov) against the same probe. The same two blocked places as U-Oct, the spawn stretch now longer, and more scatter in the outer bays and along the west wall, which this build's 10 m grid range reaches and the October build's did not.

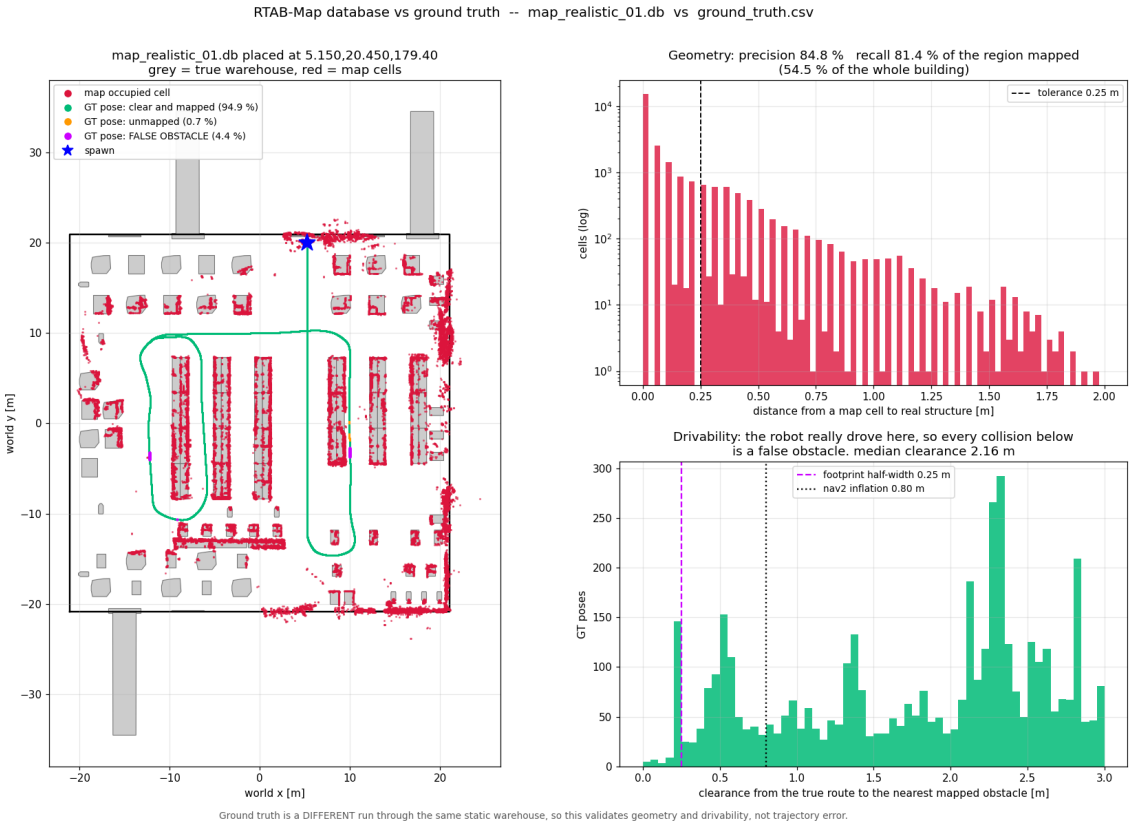


Figure 4.7: Dense-map evaluation of `map/map_realistic_01` (I-Oct) against `dataset_allsensor_realistic_001`. The clearance mode at exactly 0.25 m is the probe threading the one-aisle gap at $x \approx 10$ between two mapped shelf rows.

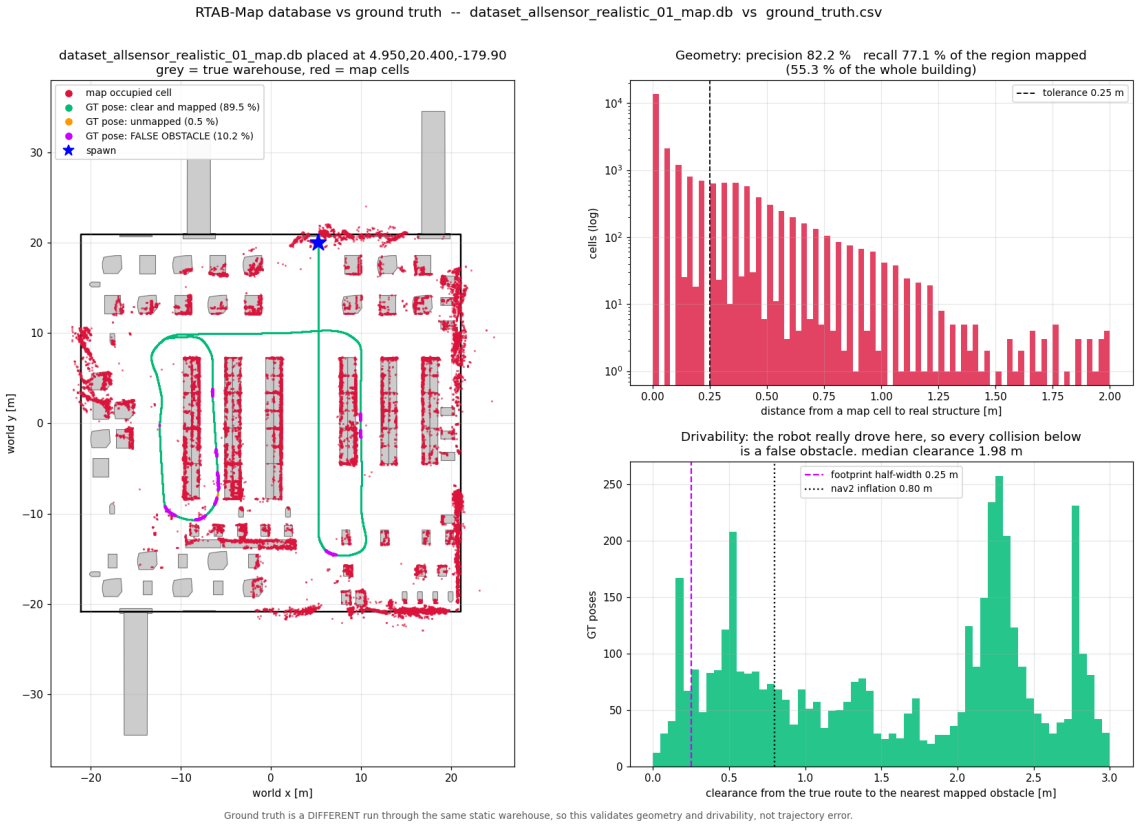


Figure 4.8: Dense-map evaluation of `map/dataset_allsensor_realistic_01_map` (I-Nov) against the same probe. The blocked poses are no longer confined to the spawn: stretches of 68, 55 and 50 poses sit mid-route in the aisles of the probe's left loop and along its bottom straight, on cells the map places on the floor of the aisle.

largest new phantom clusters sit on them. The ground-height threshold, 0.29 to 0.10 m, shows where it would: on the I layout the November build blocks 518 probe poses against 213, and the new stretches are mid-route, in the aisles, on cells first observed at median node 27 rather than at the start. A one-parameter rebuild is the test that would separate the three, and is the next action.

In the October builds the loss is mostly thickened structure, with a phantom share the tiny map did not have. Classifying the wrong cells (`script/exp4_insights.py`): 13.4 and 9.7 % of occupied cells are surfaces drawn too thick, within 0.50 m of true structure, and 5.2 and 5.3 % are phantoms further out, against 3.0 and 0.4 % on the tiny map; in the November builds the shares are 15.4 and 11.5 % thickened and 7.3 and 6.3 % phantom. The thickening sits on the shelf rows the build route passed; the phantoms are the east and south walls, drawn about 0.5 m off their true line in all four maps and far from any route. In the October builds the false obstacles are not the thickened shelving: every blocked probe pose lies within 0.6 m of a phantom cell, the largest blocked stretch of each map is the first metres from the spawn, and 96 and 63 % of the phantom cells under those stretches were first observed by one of the first five graph nodes, during the inertial initialisation transient. The November U build repeats that pattern (224 spawn poses, 88 % of the blocking cells early); the November I build does not, which is the ground-height signature above.

All four are an odometry picture by design, and should be read as one. Loop closure was disabled on both sides of the pipeline after the A/B of 26 October showed the front end's own closures corrupting an otherwise 0.1 m map (RTAB-Map record §7c), so no node was ever corrected against a revisited place; the stored graphs confirm it, with no closure spanning more than three nodes and none at all in the November builds. The thickening and the displaced walls are therefore uncorrected drift along a 365–385-metre build route re-observing the same structure from slightly different estimated places, which is what this configuration was built to expose. It is not evidence about RTAB-Map's loop closing, which was not exercised.

Whole-building recall of 54.5 to 57.1 % is route coverage, as on the tiny map: the build routes loop the central block and the outer bays were seen only in passing. Observed-region recall, the mapping figure, is 74.9 to 82.5 %.

Placement needed the overlay on three of the four fits, and the spawn check did not apply to any. The fits land about 180° from the −90° spawn prediction because the map frame was seated during the inertial initialisation transient rather than at the body's spawn pose: node 1 of the October databases carries a yaw of 57 and −89° and a height still settling over the first nodes, where the tiny map's node 1 is the identity. On the November U build the unconstrained sweep chose the rotated rival outright; the overlays leave no doubt about any of the four placements, and the two builds of each layout agree to 0.3 m and 0.5°.

Limitations, each stated once. *Not route-matched:* the tiny probe was one loop of a wide aisle, these probes thread shelf aisles (5th-percentile clearance 0.18–0.62 against 0.45 m), so part of the false-obstacle gap is probe difficulty and the 1 % threshold was set against an easier route. *No attribution:* world and route vary together between layouts, and three parameters and the front-end session vary together between builds, so no factor may be credited for any difference in this slot. *n:* one database per cell supports no uncertainty estimate. *Front end unrecorded:* the databases store the back end's parameters but not which front end, in which session, supplied the odometry chain.

[Queried: front end of the realistic builds] RTAB-Map took its pose from TF, so the odometry came from whichever front end was running; none of the four databases records which, and the November builds were made during the latency sessions of Experiment 3, whose load on the front end is not recorded either. The precision loss cannot be placed on the front end against the back end, nor the November drop on parameters against session, until the build manifest names both.

Summary. Where tracking holds, the mapping pipeline is geometrically sound on the tiny map: occupied-cell precision 96.6 % and observed-surface recall 89.5 %, with whole-building recall of 67.2 % explained by route coverage; the drivability half is a dated measurement whose independent reference is

lost. On the realistic warehouse four maps built without loop closure, two per layout under the October and the current configuration, are auditable end to end and all fail the same criterion: precision 77.8 to 84.8 % and false obstacles 3.8 to 10.2 % against independent probes. Their loss is thickened shelving, displaced walls and phantoms at the spawn from uncorrected drift, which the closure-free build was configured to expose; the current configuration scores lower on every metric, for reasons the three parameters it changed at once do not let this slot separate. The earlier large-warehouse attempt that lost tracking remains unauditable.

Future plan. Attribute the November drop by rebuilding one recording with a single parameter restored at a time, first the ground-height threshold (`Grid/MaxGroundHeight`), then the detection rate, and rescoring with the unchanged evaluator. Name the front end, its session and the three parameters in every build manifest. Use the four maps as the odometry baseline against which a corrected map is compared: re-enable closure with a translation bound on accepted corrections, or merge the closure-free sub-maps through Atlas, and rescore. Discard the cells of the first few nodes, or start the map after the inertial transient has settled, which the data predict would remove the spawn stretch. Record the tiny-warehouse independent route again and retain it.

4.4.2 Real robot

Status: planned. No RTAB-Map database has been built from real imagery, and no reference geometry of the site exists to score one against.

Reason. The geometry metric scores occupied cells against known collision geometry, and the placement step needs that geometry; the mock warehouse has no such model. The drivability metric needs a tracked route from a run other than the mapping run. Neither input exists at the cutoff.

Next action. Survey the mock warehouse (walls, stacks, walkway) into a reference occupancy grid at 0.05 m, build a database from a real mapping run, and score it with the unchanged evaluator, using the optical tracker route from a second run for drivability.